

SENTINEL-X Workflow_and_System_Architecture Document

Sentinel-X is a multi-agent autonomous AI system designed to monitor infrastructure, analyze incidents, make explainable decisions, and execute controlled recovery actions.

The system is built to demonstrate how data algorithms evolve into intelligent agents that perceive, reason, decide, and act on real-world operational data.

Project Architecture

Sentinel-X works through multiple coordinated AI agents:

1. Simulation Layer

This layer generates realistic system behavior and incidents.

Files used:

- incident_generator.py
- service_simulator.py
- test_simulation.py

Purpose:

Simulates real infrastructure conditions such as:

- service crashes
- high traffic
- unauthorized access
- connection failures

This provides realistic input data for the agents.

2. Perception Agent

File:

- perception_agent_fixed.py

Purpose:

Transforms raw incidents into structured intelligence.

What it does:

- Reads incidents.json
- Calculates severity and impact
- Generates structured incident summaries
- Saves analyzed_incidents.json

This step converts noisy logs into meaningful information.

3. Reasoning Agent

File:

- reasoning_agent_working.py

Purpose:

Uses AI to analyze incidents and provide deeper understanding.

What it does:

- Reads analyzed incidents
- Uses LLM reasoning
- Generates confidence scores
- Explains incidents in human-readable form
- Saves reasoned_incidents.json

This is the intelligence layer of the system.

4. Decision Agent

File:

- decision_agent.py

Purpose:

Applies rules, safety logic, and governance to decide what action should be taken.

What it does:

- Evaluates risk level

- Determines priority
- Generates explainable decision traces
- Saves decisions.json and decisions_enhanced.json

This ensures decisions are safe and explainable.

5. Autonomy Governor Agent

File:

- autonomy_governor.py

Purpose:

Controls how much autonomy the system should have.

What it does:

- Determines whether the system should act automatically or require human approval
- Logs governance decisions

This ensures responsible AI behavior.

6. Action Agent

File:

- action_agent.py

Purpose:

Executes recovery or mitigation steps.

Examples:

- restart services
- scale resources
- isolate components

Results saved in:

- execution_results.json

This converts decisions into real actions.

7. Root Cause Intelligence Agent

File:

- root_cause_agent.py

Purpose:

Identifies patterns across incidents to determine underlying causes rather than just reacting to symptoms.

Results saved in:

- root_cause_analysis.json

This improves long-term system intelligence.

8. Memory Layer

Folder:

memory/

Purpose:

Stores all system outputs so agents can learn and decisions remain auditable.

Files include:

- incidents.json
- analyzed_incidents.json
- reasoned_incidents.json
- decisions.json
- execution_results.json
- root_cause_analysis.json
- autonomy_governance.json

This acts as persistent memory for the system.

9. Dashboard Layer

Purpose:

Visualize:

- incidents
- reasoning
- decisions
- actions
- root causes
- system status

Streamlit will be used to connect backend data to the UI.

Key Features of Sentinel-X

- Multi-agent AI architecture
 - Explainable AI decision traces
 - Human-in-the-loop governance
 - Autonomous recovery actions
 - Root cause intelligence
 - Persistent memory
 - Realistic infrastructure simulation
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What Makes Sentinel-X Innovative

Unlike traditional monitoring systems that only display alerts, Sentinel-X:

- understands incidents
- explains reasoning
- decides safely
- acts autonomously
- learns from history

This transforms data into real-world actions.

Theme Alignment

Sentinel-X fits strongly under:

Data Algorithms in Action

As the system does not just analyze data — it uses algorithms and AI agents to make decisions and execute actions in real time.